

① • The state a substance at a particular temperature and pressure depends on two **antagonistic** entities:

- The **kinetic energy** of the particles. 对抗的
- The **strength of the attractions** between the particles.

condensed phases

凝聚相

② Intermolecular Force 分子间作用力 → 控制物理性质 (boiling, melting points, vapor pressures, viscosities) ~ 16 kJ/mol

Intramolecular Force 分子内作用力 → hold compounds together ~ 431 kJ/mol

?

③ van der Waals Forces (中性分子间)

1° Dipole-dipole interactions

得有 permanent dipoles (永久偶极) → 极性分子

↳ 负吸正, 正吸负, 靠近才起作用

more polar (极性越强) higher its boiling point

2° Dipole-Induced Dipole interactions

polar molecule → nonpolar molecule (比较弱)

3° London Dispersion Force

between an instantaneous dipole and an induced dipole (不论原本是极性还是非极性)

- The tendency of an electron cloud to

distort in this way is

called **polarizability**

极化率.

只能说原本极性的, 其更容易极化, 但不影响极化率, 还是得看大不大

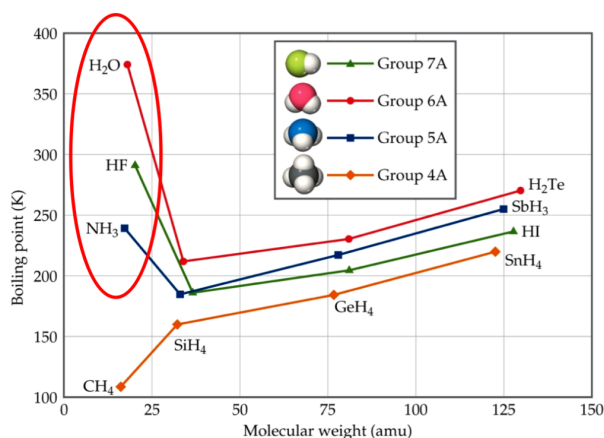
shape (long, skinny (ex n-pentane) > short, fat (neopentane)) surface area

molecular weight (越大越强, 大分子有更大电子云容易极化) 体积

Which Have a Greater Effect?

Dipole-Dipole Interactions or Dispersion Forces

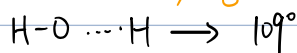
- If two molecules are of **comparable size and shape**, dipole-dipole interactions will likely be the dominating force.
- If one molecule is much larger than another, dispersion forces will likely determine its physical properties.



④ Hydrogen Bonding

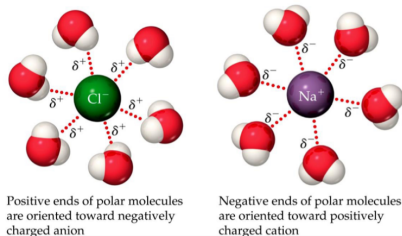
H-N.O.F

the non-hydrogen atom must have a nonbonding electron pair



⑤ Ion-Dipole Interactions

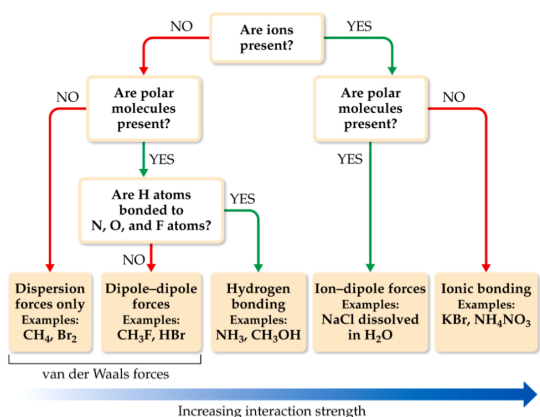
- The strength of these forces is what makes it possible for ionic substances to dissolve in polar solvents.



解离后的离子和水之间

↳ 极性溶剂

Summarizing Intermolecular Forces



总结为：先看是否离子化合物
再看有无氢键，再看相对分子质量

⑥ 液体的一些性质

1° Viscosity 粘度

定义：Resistance of a liquid to flow is called viscosity

随分子间作用力 ↑ 而 ↑

T ↑ 而 ↓

2° surface Tension 表面张力

随分子间作用力 ↑ 而 ↑

T ↑ 而 ↓

3° Capillary Action 毛细作用

分子间作用力 $\left\{ \begin{array}{l} \text{自己和自己的} \rightarrow \text{cohesive} \\ \text{自己和其它的} \rightarrow \text{adhesive} \end{array} \right.$

downward U

Wetting 浸润 (adhesive > cohesive — H₂O - glass > H₂O - H₂O)

Inverted U

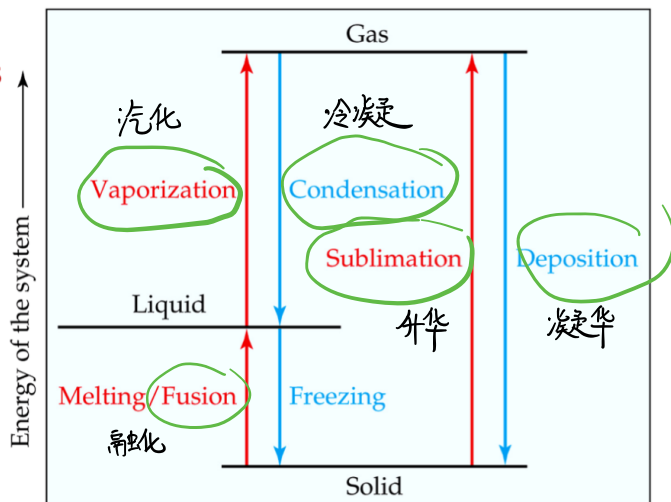
Non-Wetting 非浸润 (adhesive < cohesive — Hg - glass < Hg - Hg)

玻璃换成 Wax

水变为不浸润，Hg 不变

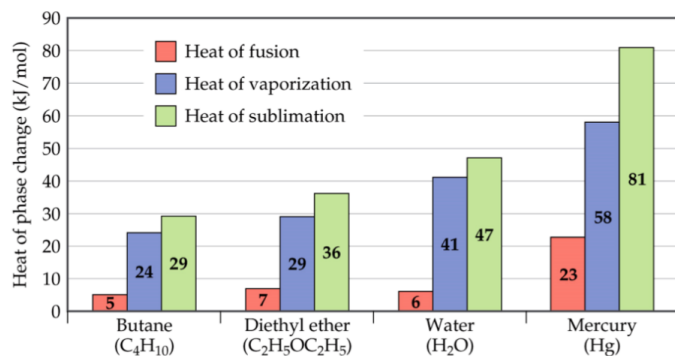
① phase change

Phase Changes



— Endothermic process (energy added to substance)
— Exothermic process (energy released from substance)

Energy Changes Associated with Changes of State



The **heat of fusion** is the energy required to change a solid at its melting point to a liquid.

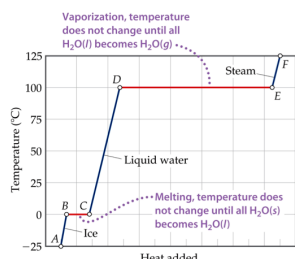
The **heat of vaporization** is defined as the energy required to change a liquid at its boiling point to a gas.

都是温度不变状态了

The **heat of sublimation** is defined as the energy required to change a solid directly to a gas.

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$$

Energy Changes Associated with Changes of State



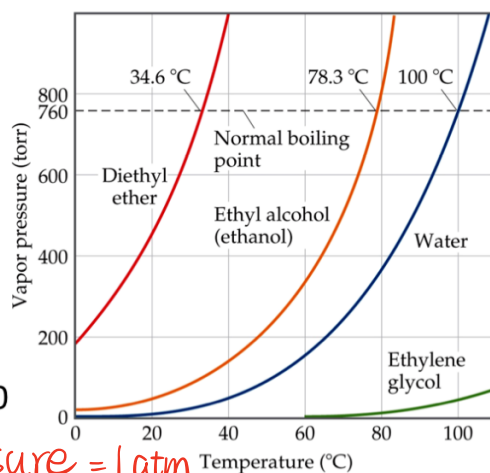
- The heat added to the system at the melting and boiling points goes into pulling the molecules farther apart from each other.
- The temperature of the substance does not rise during a phase change.

相变的过程中温度保持不变

⑧ Vapor Pressure

Vapor Pressure

- The **boiling point** of a liquid is the temperature at which its **vapor pressure equals atmospheric pressure**.
- The **normal boiling point** is the temperature at which its vapor pressure is 760 torr.

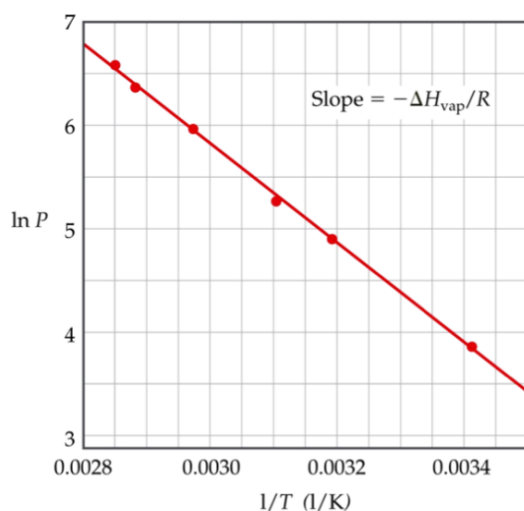


vapor pressure = 1 atm

分子间作用力 ↑



Vapor Pressure



The natural log of the vapor pressure of a liquid is inversely proportional to its temperature.

T ↑ P ↑

This relationship is quantified in the **Clausius–Clapeyron equation**:

$$\ln P = -\frac{\Delta H_{\text{vap}}}{RT} + C,$$

where C is a constant.

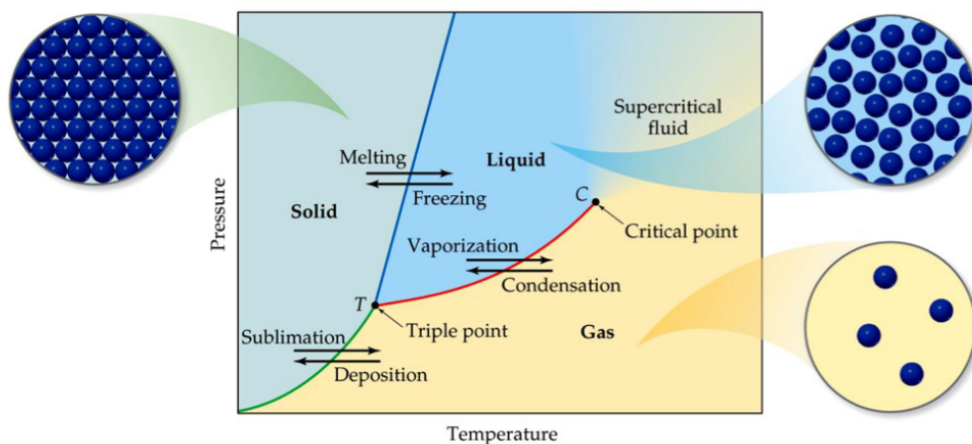
and $R = 8.314 \text{ J/mol}\cdot\text{K}$.

- Large intermolecular forces in a substance are shown by
- A) low vapor pressure
 - B) high boiling point
 - C) high heats of fusion and vaporization
 - D) high critical temperatures and pressures
 - E) all of the above

critical temperature and pressure

- The highest temperature at which a distinct liquid phase can form is called the **critical temperature**. (eg. Water: 374 °C)
- The **critical pressure** is the pressure required to bring about liquefaction at its critical temperature. (eg. Water: 217.7 atm)

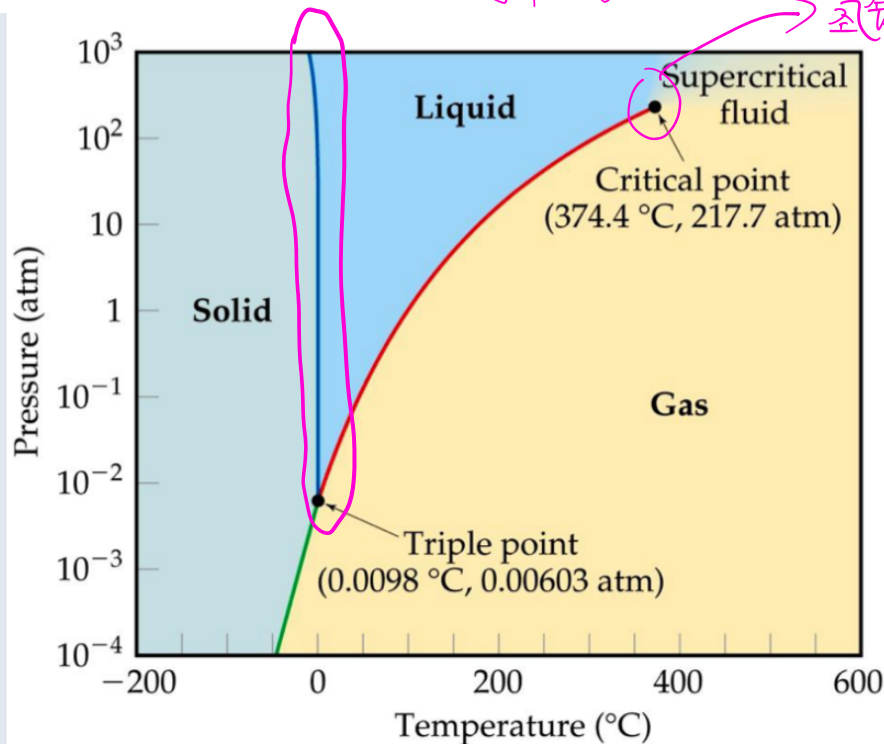
⑨ Phase Diagrams



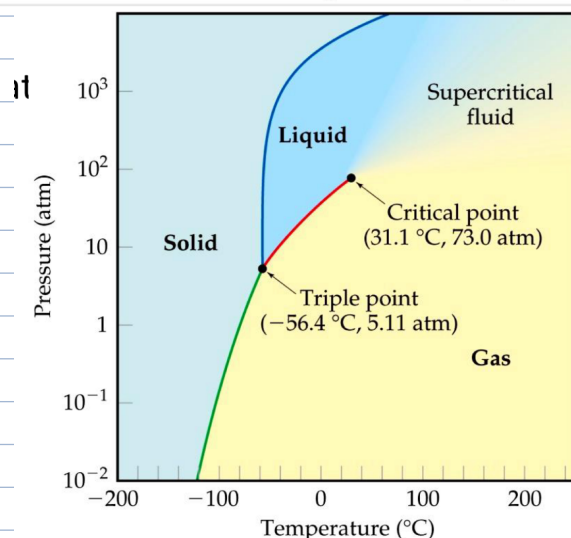
- Below the triple point the substance cannot exist in the liquid state.

水的相图

负的 (同 T P) $s \rightarrow L$ $\rightarrow$ 冰的密度比水小 (反常)
 $\rightarrow$ 氢键



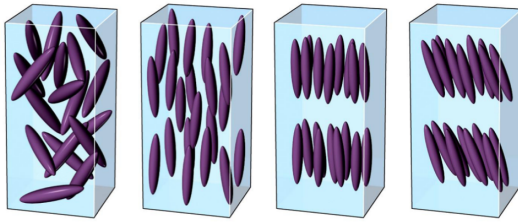
CO₂的相图



CO₂ sublimates at normal pressure

⑩ Liquid Crystals

Liquid Crystals



Liquid phase
Molecules arranged randomly

Nematic liquid crystalline phase

Long axes of molecules aligned but ends are not aligned

Smectic A liquid crystalline phase

Molecules aligned in layers, long axes of molecules perpendicular to layer planes

Smectic C liquid crystalline phase

Molecules aligned in layers, long axes of molecules inclined with respect to layer planes

排整齐

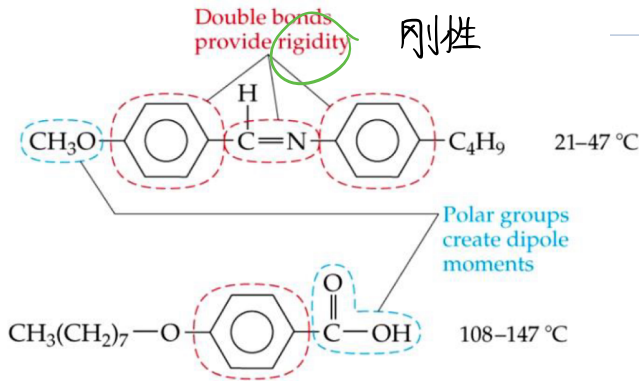
Unlike liquids, molecules in liquid crystals have some degree of order.

有三种
Nematic
Smectic
cholesteric

14. Molecules with only single bonds do not generally exhibit liquid-crystalline properties because _____.

- A) molecules with only single bonds are too big to exhibit liquid-crystalline properties
- B) molecules without multiple bonds lack the flexibility necessary for alignment
- C) molecules with single bonds are gases
- D) molecules without multiple bonds lack the rigidity necessary for alignment
- E) molecules without multiple bonds are too small to exhibit liquid-crystalline properties

单键 → 缺乏刚性 → 旋转
2/4 刚性 利于 液晶



color change ← temperature change

Isopropyl alcohol

异丙醇

高海拔、大气压下降